

src/app/BackgroundActivityContext.tsx

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Overview

A new `src/app/BackgroundActivityContext.tsx` file introduces a React context for tracking background tasks, exposing a provider and two hooks.

Key changes

- **Imports:** Adds `createContext`, `useCallback`, `useContext`, `useMemo`, `useState`, and `ReactNode` from `react` (lines 1-6).
- **Types:** Declares `BackgroundActivityKind` enum-like union (lines 10-19) and `BackgroundActivity` interface (lines 22-30).
- **Context value:** Defines `BackgroundActivityContextValue` (lines 37-41) and creates the context (lines 44-46).
- **Provider:** Implements `BackgroundActivityProvider` (lines 48-95) with state, upsert, remove, and clear logic, sorting activities by `startedAt`.
- **Hooks:** Exposes `useBackgroundActivity` (lines 97-103) that throws if used outside the provider, and `useBackgroundActivityOptional` (lines 106-108) that returns `null` when no provider is present.

Impact

- **Correctness:** The provider guarantees a sorted activity list; the optional hook prevents crashes in optional embed paths.
- **Maintainability:** Centralized activity state simplifies adding new activity kinds or UI components.
- **Performance:** Sorting on every upsert may be costly with many activities; current implementation is acceptable for typical use.
- **Compatibility:** No changes to existing modules; the new file is self-contained and only adds exports.

Risks & follow-ups

- **Provider usage:** Verify that all components consuming the hooks are wrapped by `BackgroundActivityProvider`.
- **Null handling:** `useBackgroundActivityOptional` may return `null`; callers must guard against it.
- **Sorting overhead:** Benchmark with high activity churn to ensure sorting does not become a bottleneck.
- **Default values:** The provider defaults `kind` to `"other"`; confirm this aligns with intended UX for unknown kinds.